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Predicting the appearance of a user following treatment

Abstract

According to an aspect, there is provided an apparatus comprising a processor configured to receive image data in respect of a user (**102**) of a personal care device (**104**), wherein the personal care device is to be used to treat a portion of the user; receive device operating data relating to the personal care device; determine position information of at least a treatment element (**112**) of the personal care device relative to the user; generate, based on the received image data, the received device operating data and the determined position information, an image (**120**) representative of a predicted appearance of the portion of the user following treatment using the personal care device; and provide the generated image for display on a representation of the user.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1025131	12/1911	Donnell	N/A	N/A
10259131	12/2018	Krenik	N/A	N/A
2015/0055085	12/2014	Fonte et al.	N/A	N/A
2015/0321365	12/2014	Lauritsen	N/A	N/A
2016/0262521	12/2015	Kustra et al.	N/A	N/A
2019/0053607	12/2018	Shen	N/A	G06F 3/04842
2019/0152075	12/2018	Hoexum	N/A	A45D 27/42
2020/0069034	12/2019	Godlieb	N/A	G06T 11/00
2020/0316799	12/2019	Tsegenidis	N/A	B26B 21/4087

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108021044	12/2017	CN	N/A
3366195	12/2017	EP	N/A
2018007547	12/2017	WO	N/A
2018104356	12/2017	WO	N/A
WO-2018149738	12/2017	WO	B26B 19/20
2019001894	12/2018	WO	N/A

OTHER PUBLICATIONS

Wu et al., “Look at Boundary: A Boundary-Aware Face Alignment Algorithm”, Tsinghua National Laboratory for Information Science and Technology (TNList), Department of Computer Science and Technology, Tsinghua University Sense Time Research, Amazon Rekognition, May 26, 2018. cited by applicant

Hawkins et al., “Assistive Mobile Manipulation for Self-Care Tasks Around the Head”. cited by applicant

International Search report and Written Opinion of PCT/EP2021/064732 dated Oct. 15, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATIONS

(1) This application is the U.S. National Phase application under 35 U.S.C. § 371 of International Application No. PCT/EP2021/064732, filed on Jun. 2, 2021, which claims the benefit of European Patent Application No. 20178895.7, filed on Jun. 9, 2020. These applications are hereby incorporated by reference herein.

FIELD OF THE INVENTION

(2) The invention relates to personal care and, more particularly, to predicting the appearance of a user of a personal care device following use of the device.

BACKGROUND OF THE INVENTION

(3) When performing a personal care activity, such as shaving or trimming hair, it is often desirable to perform the particular activity accurately in order to achieve the desired result. For example, when trimming a beard, a user may intend that the beard is trimmed similarly on both sides of the face, and may intend for the beard to appear symmetrical.

(4) To aid the user, the user may perform the personal care activity in front of a mirror, so that they can view their reflection, or in front of a device capable of displaying a live image of the user while the activity is performed. Even then, it may be difficult to know what the result will be of performing the personal care activity.

SUMMARY OF THE INVENTION

(5) In contrast to existing systems, embodiments of this disclosure provide a mechanism by which a user can see a visual prediction of an outcome of a personal care activity before the activity has been performed. The inventors have recognised that, by using various pieces of data, it is possible to generate an image illustrating an effect of using a personal care device, such that a user can see the effect of the device before it is used. For example, a user of a shaving device may be able to view a representation of their face which includes a modification to illustrate how the user's face might look if the shaving device is applied to a particular portion of their face. In this way, the user is able to make a decision regarding whether or not to proceed with the treatment (e.g. shaving procedure) before any irreversible changes have been made. Moreover, by displaying the predicted modification on a representation of the user's face at a position based on the position of the personal care device, the user may be able to position the device more accurately in an intended position, thereby making easier for the user to perform personal care activity as intended.

(6) According to a first specific aspect, there is provided an apparatus comprising a processor configured to receive image data in respect of a user of a personal care device, wherein the personal care device is to be used to treat a portion of the user; receive device operating data relating to the personal care device; determine position information of at least a treatment element of the personal

care device relative to the user; generate, based on the received image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; and provide the generated image for display on a representation of the user.

(7) According to some embodiments, the processor may be further configured to determine, based on the received image data, a degree of deformation of a surface of the user by the personal care device. The processor may be further configured to generate the image representative of the predicted appearance of the portion of the user further based on the determined degree of deformation of the surface of the user.

(8) The processor may, in some embodiments, be configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of a sensor of the personal care device.

(9) In some examples, the device operating data may comprise a size of the treatment element of the personal care device. The processor may be configured to generate the image representative of the predicted appearance of the portion of the user having a size based on the size of the treatment element.

(10) The device operating data may comprise data indicative of an orientation and/or a motion of the personal care device. The processor may be configured to generate the image representative of the predicted appearance of the portion of the user having a size based on the orientation/motion data.

(11) In some examples, the processor may be further configured to determine, based on the received image data, an indication of the body part of the user that is to be treated using the personal care device. The processor may be further configured to generate the image representative of the predicted appearance of the portion of the user having a size based further on the indication of the body part.

(12) In some embodiments, the apparatus may comprise an interactive mirror, a desktop computer, a laptop computer, a tablet computer, a smartphone or a wearable device.

(13) According to a second aspect, there is provided a computer-implemented method comprising receiving image data in respect of a user of a personal care device, wherein the personal care device is to be used to treat a portion of the user; receiving device operating data relating to the personal care device; determining position information of at least a treatment element of the personal care device relative to the user; generating, based on the received image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; and displaying the generated image on a representation of the user.

(14) The computer-implemented method may further comprise displaying the representation of the user and the generated image on the representation of the user in an alternating manner.

(15) In some embodiments, the computer-implemented method may further comprise determining, based on the received image data, an amount of deformation of a surface of the user by the personal care device. Generating the image representative of the predicted appearance of the portion of the user may be further based on the determined amount of deformation of the surface of the user.

(16) According to third aspect, there is provided a computer program product comprising a non-transitory computer-readable medium, the computer-readable medium having computer-readable code embodied therein, the computer-readable code being configured such that, on execution by a suitable computer or processor, the computer or processor is caused to perform steps of the methods disclosed herein.

(17) According to a fourth aspect, there is provided a personal care system comprising a personal care device having a treatment element to be used to treat a portion of a user; an image capture device configured to capture image data in respect of the user of the personal care device; a display unit; and processing apparatus configured to operate the image capture device to capture image

data in respect of the user of the personal care device; receive device operating data relating to the personal care device; determine position information of at least the treatment element of the personal care device relative to the user; generate, based on the captured image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; and provide the generated image for display on a representation of the user on the display unit.

(18) In some examples, the personal care device may comprise a first sensor configured to measure a force or pressure applied to the user by the personal care device. The processing apparatus may be configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of the first sensor.

(19) The personal care device may, in some in bottoms, comprise a second sensor configured to measure orientation data and/or motion data in respect of the personal care device. The processing apparatus may be configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of the second sensor.

(20) In some examples, the personal care device may comprise a device selected from a group comprising: a hair trimming device, a shaving device, a hair treatment device, a photoepilation device, a skin treatment device, a skin rejuvenation device and a product application device.

(21) These and other aspects will be apparent from and elucidated with reference to the embodiment(s) described hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments will now be described, by way of example only, with reference to the following drawings, in which:

(2) FIG. 1 is an illustration of user performing a personal care activity;

(3) FIG. 2 is a schematic illustration of an example of an apparatus according to various embodiments;

(4) FIGS. 3A, 3B and 3C are illustrations of images generated according to various embodiments;

(5) FIG. 4 is an illustration of images generated according to various embodiments;

(6) FIGS. 5A and 5B are illustrations of images showing various embodiments of the invention in use;

(7) FIG. 6 is a flowchart of an example of a method according to various embodiments;

(8) FIG. 7 is a flowchart of a further example of a method according to various embodiments;

(9) FIG. 8 is a schematic illustration of a processor in communication with a computer-readable medium; and

(10) FIG. 9 is a schematic illustration of a personal care system.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) According to various embodiments of the invention, methods, apparatuses, systems and computer program products are provided that enable a user of a personal care device to view a visual indication of the effect of performing the personal care activity before the activity has been performed. Various embodiments are described in the context of personal care activities performed using personal care devices, such as haircare devices and skin care devices. However, it will be appreciated that the disclosed embodiments may be implemented using any type of personal care device.

(12) Referring to the drawings, FIG. 1 is an illustration of a user **102** of a personal care device **104**. In this example, the personal care device **104** comprises a shaving device of the type used to shave hair, for example from the face of the user **102**. In some cases, particularly when the user **102** cannot directly view the portion of their body being treated, the user may perform personal care

activities (e.g. shaving) in front of a mirror so that they can position the personal care device **104** more accurately, to make it easier to achieve the desired outcome. In the example shown in FIG. 1, the user **102** is performing the personal care activity in front of a display device **106** capable of displaying a representation of at least part of the user, for example a reflection of the user or a representation of the user that has been captured using a camera.

(13) In some examples, the display device **106** may comprise an interactive mirror. An interactive mirror, sometimes referred to as a smart mirror, is a unit which, in addition to functioning as a mirror to show a user his or her reflection, is also capable of displaying information to the user. Information, such as text, images and videos, may be displayed on a display portion of the interactive mirror which may, for example, be positioned behind a mirrored (or partially mirrored) panel or a mirrored (or partially mirrored) surface. In this way, the display screen, or portions thereof, may be visible through the mirror portion, so that a user is able to simultaneously view their reflection and information presented on the display screen. The interactive mirror **106** may include one or more sensors, such as an imaging device **108** (e.g. a camera) capable of capturing an image (e.g. of the user **102**) which can be displayed on the screen of the interactive mirror or used for determining data (e.g. device position data) as explained below. The interactive mirror **106** may be connected (wirelessly or via a wired connection) to one or more other devices, such as the personal care device **104**. The processing circuitry within the interactive mirror **106** may receive data from the personal care device **104** (and/or from other connected devices) and may display information to the user relating to the connected device.

(14) In other examples, an alternative display device **106** may be used by the user **102** instead of an interactive mirror. For example, the user **102** may perform the personal care activity in front of a smart phone, a tablet computer, a laptop computer, or some other display device capable of displaying a representation of the user along with other information. For example, the camera **108** may form part of the display device or may be a separate device that is connected to the display device **106**.

(15) The personal care device **104** may comprise a body **110** and a treatment element **112** for performing the intended treatment of the personal care device. For example, in a shaving device, the treatment element **112** may comprise a shaving head having blades that cut hair during use. The body **110** of the personal care device **104** may house various components. In some examples, the personal care device **104** may include one or more sensors, such as a first sensor **114** and/or a second sensor **116** for measuring data relating to the personal care device and/or its use.

(16) Various steps of the methods disclosed herein may be performed using one or more processors, such as a processor **118**, which may form part of a computing device or cloud computing environment connected to the display device **106**. In examples where the display device **106** comprises part of a computing device (e.g. a smartphone or tablet computer), the processor **118** may also form part of the computing device. In the example shown in FIG. 1, the processor **118** forms part of the interactive mirror **106**. According to embodiments of the invention, during use of the personal care device **104**, the processor **118** may use various data (e.g. image data acquired using the camera **108** and/or data acquired using the sensors **114**, **116**) to generate an image **120** that represents the predicted appearance of part of the user **102** following treatment using the personal care device **104**. The generated image **120** may comprise a patch that can be displayed on the display device **106** with the representation (e.g. the reflection or the generated representation) of the user **102**. For example, the generated image **120** may be presented as an overlay or a superimposed region on the display device **106**. In this way, the user **102** is able to view themselves (i.e. the representation of them) and, at the same time, see a prediction of what they will look like if they continue to use the personal care device **104** in a particular region.

(17) In some embodiments, the generated image **120** may be displayed on the display device **106** as the personal care device **104** approaches the user **102** (e.g. when the personal care device is within a defined threshold distance of the user) but before the treatment element **112** of the personal care

device treats the user. Moreover, the position of the generated image **120** relative to the representation of the user **102** on the display device **106** may be based on the position of the personal care device **104** or the position of the treatment element **112** that the personal care device, such that movement of the treatment element **112** results in movement of the image **120** on the display device. Thus, prior to performing the personal care activity, which may result in irreversible changes being made, the user **102** may view the predicted outcome shown by the generated image **120** and decide whether or not to proceed with the personal care activity.

(18) According to a first aspect, the present invention provides an apparatus. FIG. 2 is a schematic illustration of an example of an apparatus **200** may be used in the implementation of embodiments of the invention. The apparatus **200** comprises a processor **202**. The processor **202** may comprise or be functionally similar to the processor **118**. The apparatus **200** may comprise or form part of a computing device such as a smartphone or a tablet computer, or may comprise or form part of an interactive mirror. The processor **202** is configured to receive image data in respect of a user **102** of a personal care device **104**, wherein the personal care device is to be used to treat a portion of the user. The image data may, for example, be acquired from an image capture device or camera **108** forming part of a computing device or interactive mirror, or from an image capture device or camera that is separate from, but capable of delivering image data to, the apparatus **200**. The image data, which may be captured using known image capture techniques, in any known image format, may include part of the user **102** and, in some examples, may also include some or all of the personal care device **104**. In examples where an image of at least part of the personal care device **104** is captured in addition to an image of at least part of the user **102**, it may be possible to determine information from the image data, such as data indicative of relative position, orientation and/or movement of the personal care device **104** and the user **102**.

(19) The camera **108** may comprise a 2D or 3D camera (e.g. a stereo camera) and may be capable of using techniques such as time of flight measurement. Data acquired using the camera **108** may be used to create a 2D and/or 3D model of at least part of the user **102**. Determination of the location and/or orientation of the personal care device **104** relative to the user **102** (sometimes referred to as device tracking) may be achieved by monitoring the position of the user in the captured image. When the user's face is captured in the image data, the face position and orientation may be detected using facial landmarks (e.g. points of the eyes, nose mouth, and so on). The positions of the landmarks are then incorporated onto a pre-recorded 3D facial model of the user **102**. The position and/or orientation of the personal care device **104** may be obtained by detecting landmarks on the personal care device (e.g. reference points or corners of the device), sometimes referred to as device landmarks. These device landmarks may be inherent of the device aspect or may be specifically integrated into the personal care device to improve the tracking ability (e.g. green dots at specific positions on the device). The device landmarks may be analysed with respect to the facial landmarks of the user **102** to provide data indicative of the position and/or orientation of the personal care device relative to the portion of the user **102** (e.g. the face).

(20) The processor **202** is also configured to receive device operating data relating to the personal care device **104**. Device operating data may comprise data indicative of the type or nature of the personal care device **104** being used, and/or details of the treatment element **112** being used on the personal care device. For example, device identification data may indicate the type of personal care device being used, and/or parameter information may indicate settings (e.g. cutting length). In some embodiments, device operating data may include data acquired using one or more sensors associated with the personal care device **104**. For example, device operating data may be acquired from a sensor or sensors located within the personal care device **104** or from a sensor or sensors external to but associated with the personal care device. In other examples, device operating data may include data relating to a position, orientation and/or movement of the device relative to a reference, such as the apparatus **200** or the user **102**. Thus, the device operating data may be data that has been stored in respect of the personal care device **104** prior to the use of the device or data

that is acquired in real-time during the use of the personal care device.

(21) The device operating data may be received from the personal care device **104** itself or from a storage (e.g. a memory) associated with the processor **202**, in which such data is stored. For example, the memory may store a database containing device operating data in respect of several personal care devices **104**.

(22) The processor **202** is further configured to determine position information of at least a treatment element **112** of the personal care device **104** relative to the user **102**. Position information may include a location or position of at least the treatment element **112** relative to a reference point or object, such as a portion of the user **102** or the apparatus **200**, an orientation of at least the treatment element relative to the reference point or object, a distance from at least part of the treatment element to the reference point or object and/or an indication of a direction of motion of at least the treatment element relative to the reference point or object. The position information may be determined from the image data received in respect of user of the personal care device **104** and/or from one or more other sources, such as the device operating data and sensors within the personal care device.

(23) The processor **202** is further configured to generate, based on the received image data, the received device operating data and the determined position information, an image **120** representative of a predicted appearance of the portion of the user **102** following treatment using the personal care device **104**. The processor **202** is further configured to provide the generated image for display on a representation of the user **102**. For example, the image **120** and the representation of the user may be presented on the display **106**. In examples where the display **106** forms part of a computing device, such as a smartphone or tablet computer, the representation of the user **102** may be an image or live video footage of the user (i.e. image data) acquired using the image capture device **108**. In examples where the display **106** forms part of an interactive mirror, then the representation of the user **102** may be a reflection of the user in the mirror.

(24) The image **120** representing the predicted appearance of the portion of the user **102** may generally be smaller than the display **106** on which the representation of the user **102** is presented, and may be sized so that it is smaller than the representation of the user. The image **120** may be considered to be a patch or mask (e.g. square or rectangular in shape) that is intended to cover or replace a portion of the representation of the user. For example, if the user **102** is performing a shaving activity to shave one of their sideburns, then the image **120** generated by the processor **202** may comprise a patch (e.g. rectangular) representing the appearance of a portion of the side of the user's face that is intended to be shaved using the personal care device **110** (e.g. a shaver). The image **120** may comprise a representation of the user's skin with no hair or shorter hair, and this may be presented to the user **102** as a mask, overlay or otherwise superimposed on the representation of the user on the display **106**.

(25) To generate the image representative of the predicted appearance of the portion of the user **102** following treatment, the processor **202** may, in some embodiments, use prestored data (e.g. data stored in a memory) indicating or describing how a particular portion of the user might appear if the personal care activity is carried out. For instance, in the shaving examples, the generated image **120** may comprise a stored image of clean-shaven skin, so that the user **102** can see what a particular portion of their body might look like clean-shaven. A more accurate prediction of the appearance of the portion of the user may be made using additional data available to the processor **202**. For example, data indicative of the user's skin color, tone, complexion and the like may be determined from the received image data, and the image **120** of the predicted appearance of the portion of the user **102** may be generated using such data, so that the generated image is more realistic.

(26) In some examples, device operating data relating to the personal care device (e.g. the shaver) **110** may be used to improve the prediction. FIG. 3 is an illustration of three different examples of images **120** generated using device operating data relating to the shaver. In each example, a

sideburn of a user **102** is shown with part of the user's hair. In FIG. 3A, device operating data of the shaver may indicate that the treatment element **112** of the shaver is set to perform a clean-shaven (e.g. with no shaving guide in place). In this example, therefore, the image **120** may show a clean-shaven portion of the user's skin to replace the sideburn. In FIG. 3B, device operating data of the shaver may indicate that the treatment element **112** of the shaver is set to cut the user's hair very short and, therefore, the image **120** may show a portion of the user's skin with stubble replacing the sideburn. In FIG. 3C, device operating data of the shaver may indicate that the treatment element **112** of the shaver is set merely to trim the user's hair and, therefore, the image **120** may show a portion of the user's skin with hair that is slightly shorter than the user's current hair length in the sideburn region. In some examples, the device operating data indicating the nature of the treatment element **112** may be acquired automatically from the personal care device **104** while, in other examples, a user may provide (e.g. via user interface, not shown) details of the treatment element, such as an intended length of cut.

(27) An appearance (e.g. a size, shape and/or orientation) of the image **120** generated by the processor **202** may be based on data available to the processor **202** including, for example, the received image data, the received device operating data and/or the determined position information. FIG. 4 is an illustration of two different examples of the appearance of the image **120**. In a first example, labelled **402**, the generated image **120** is relatively small, covering a small proportion of the representation of the user **102**. In a second example, labelled **404**, the generated image **120** is relatively larger, covering a larger proportion of the representation of the user **102**.

(28) The processor **202** may determine from image data (e.g. from the camera **108**) that the personal care device **104** is to be used to shave a sideburn of the user **102** in example **402** and used to shave a cheek of the user in example **404**. In some examples, if a relatively smaller portion of the user is to be treated using the personal care device **104**, then the generated image **120** may be relatively smaller (example **402**) while if a relatively larger portion of the user is to be treated, then the generated image may be relatively larger (example **404**). Device operating data relating to the personal care device **104** may indicate an orientation in which the personal care device **104** is being held relative to the head of the user **102**. The orientation of the image **120** generated by the processor **202** may, in some examples, correspond to the orientation of at least part of the personal care device **104**, such as the treatment element **112**. Device operating data and/or position information of the treatment element **112** and/or the personal care device **104** may provide an indication of a speed of movement of the personal care device relative to the user **102**. For example, when shaving a relatively smaller region such as a sideburn, the user **102** may move the personal care device **104** slower than when shaving a larger portion such as a cheek or a neck. In other examples, the received image data may be used to determine which part of the user **102** is to be treated using the personal care device **104**, and this may be used to determine the size of the image **120**. For example, the processor **202** may be configured to determine, based on the received image data, an indication of the body part of the user **102** that is to be treated using the personal care device **104**. The processor **202** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** having a size based further on the indication of the body part.

(29) When the personal care device **104** is determined to be moving relatively quickly, the generated image **120** may be relatively larger than when the personal care device is determined to be moving relatively slowly. Thus, the device operating data may comprise data indicative of an orientation and/or a motion of the personal care device **104**. The processor **202** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** having a size based on the orientation/motion data. Various limits and thresholds may be defined and used by the processor **202** when determining the appearance of the image **120**.

(30) In some embodiments, the size of the image **120** may correspond to, or be based on details of the treatment element **112** of the personal care device **104**. For example, the device operating data

may comprise a size of the treatment element **112** of the personal care device **104**. The processor **202** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** having a size based on the size of the treatment element **112**.

(31) The processor **202** may, in some embodiments, be configured to determine, based on the received image data, a degree of deformation of a surface of the user **102** by the personal care device **104**. For example, some personal care devices may be used by touching a treatment element **112** of the personal care device **104** against the skin of the user **102** or pushing the treatment element against the skin of the user. Contact between the treatment element **112** (or any other part of personal care device **104**) and a surface (e.g. the skin) of the user **102** may cause the surface to be deformed. Using the received image data (e.g. image data acquired using the camera **108**), an amount or degree of deformation may be measured or determined, and this may be taken into account when generating the image **120**. The image data may be used, for example, to determine a depth of the personal care device **104** in the 3D model of the user **102**. In some embodiments, parameter information relating to the surface of the user may be used to improve the determination of the amount of deformation. For example, skin elasticity values at different positions on the user's body may be used. Known values of skin elasticity may be stored in a memory accessible to the processor **202**, for example. In some examples, data indicating an ethnicity and/or an age of the user may be used to determine an amount of deformation of the skin. Thus, the processor **202** may be further configured to generate the image representative of the predicted appearance of the portion of the user **102** further based on the determined degree of deformation of the surface of the user. For example, if the received image data indicates that the skin of the user **102** has been deformed by the personal care device **104**, then the image **120** generated by the processor **202** may show such deformation.

(32) In other examples, deformation of the skin of the user **102** resulting from the personal care device **104** may be determined or measured in another way. For example, the first sensor **114** and/or the second sensor **116** may comprise a force sensor or a pressure sensor to measure an amount of force or pressure applied by the personal care device **104** onto a surface of the user **102**. The data acquired using the sensor's **114**, **116** may be used when generating the image **120**. For example, an indication of an amount of force applied onto the skin of the user **102** may be used to predict an amount of deformation of the skin, and this deformation may be displayed in the image **120**. Thus, the processor **202** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** further based on an output of a sensor **114**, **116** of the personal care device **104**.

(33) In some examples, the image **120** may be displayed on the representation of the user (e.g. on the display **106**) only when certain conditions are met. For example, the image **120** may be displayed only if the personal care device **104** is determined to be within a defined threshold distance of the user **102**. FIG. 5 is an illustration of two examples of representations of the user **102** and the personal care device **104** according to embodiments of the invention. In FIG. 5A, data available to the processor **202** (e.g. received image data, received device operating data and/or determined position information) may indicate that the personal care device **104** is beyond a threshold distance from the user **102**. In this example, the image **120** may not be displayed on the representation of the user **102**. In FIG. 5B, however, the data available to the processor **202** may indicate that the personal care device **104** is within a threshold distance from the user **102**. In this example, the image **120** is displayed on the representation of the user **102**. By displaying the image **120** only when the personal care device **104** is within a defined threshold distance of the user **102**, the processor **202** may be able to determine an intended treatment location of the user to higher confidence level. Moreover, the user **102** may have some control over when the image **120** is to be displayed on the display **106**. For example, if the user **102** wishes to view the representation without the image **120** overlay, then they may simply move the personal care device **104** away from their face.

(34) As noted briefly above, the apparatus **200** may comprise or form part of a computing device. In some examples, the apparatus **200** may comprise an interactive mirror, a desktop computer, a laptop computer, a tablet computer, a smartphone or a wearable device (e.g. a smart watch).

(35) According to a second aspect, the present invention provides a method. FIG. **6** is a flowchart of an example of a method **600**. The method **600**, which may comprise a computer-implemented method, may be considered to be a method for predicting an appearance of the user following treatment using a personal care device. The method **600** comprises, at step **602**, receiving image data in respect of a user **102** of a personal care device **104**, wherein the personal care device is to be used to treat a portion of the user. At step **604**, the method **600** comprises receiving device operating data relating to the personal care device **104**. The method comprises, at step **606**, determining position information of at least a treatment element **112** of the personal care device **104** relative to the user **102**. At step **608**, the method comprises generating, based on the received image data, the received device operating data and the determined position information, an image **120** representative of a predicted appearance of the portion of the user **102** following treatment using the personal care device **104**. The method **600** comprises, at step **610**, displaying the generated image **120** on a representation of the user **102**. For example, the image **120** may be displayed on the display **106**.

(36) FIG. **7** is a flowchart of a further example of a method **700**. The method **700** may comprise a computer implemented method and may include one more steps of the method **600**. The method **700** may comprise, at step **702**, displaying the representation of the user **102** and the generated image **120** on the representation of the user in an alternating manner. Thus, the image **120** may be periodically displayed over the representation of the user **102** and removed, such that the image **120** repeatedly appears for a short period of time (e.g. 0.5 seconds or 1 second). In this way, the user **102** is able to compare how they currently look with how they might look if they proceed with the treatment using the personal care device **104**.

(37) In other embodiments, the display and/or removal from display of the image **120** may be controlled by the user **102**. For example, the user **102** may perform a defined movement or gesture (e.g. using a finger, a hand, their head, a facial expression, or a defined movement of the personal care device) which may be captured using the camera **108** and interpreted as an indication that the user wishes to display or remove from display the image **120**. In another example, a pause in movement of the personal care device **104** by the user **102** may be detected and recognised as a signal to display the image **120**. In this way, a user may stop moving the personal care device **104** when a would like to understand what effect further movement of the device might have. In other embodiments, the display and/or removal from display of the image **120** may be effected via a user interaction with a user interface of the personal care device **104** or the computing device (e.g. the interactive mirror or tablet computer) housing the display **106**. For example, the user **102** may press a button to switch between cases where the image **120** is displayed and not displayed.

(38) At step **704**, the method **700** may comprise determining, based on the received image data, an amount of deformation of a surface of the user **102** by the personal care device **104**. For example, contact between the personal care device **104** and skin the user **102** may cause deformation of the skin, and the amount and nature of the deformation may be determined. Generating (step **608**) the image **120** representative of the predicted appearance of the portion of the user **102** may be further based on the determined amount of deformation of the surface of the user. Thus, the image **120** generated at step **608** may include deformation of the surface.

(39) One or more steps of the methods **600**, **700** may be performed using the processor **202** of the apparatus **200**.

(40) According to a third aspect, the present invention provides a computer program product. FIG. **8** is a schematic illustration of an example of a processor **802** in communication with a computer-readable medium **804**. According to various bodies, a computer program product may comprise a non-transitory computer-readable medium **804**, the computer-readable medium having computer-

readable code embodied therein, the computer-readable code being configured such that, on execution by a suitable computer or processor **802**, the computer or processor is caused to perform steps of the methods and **600**, **700** disclosed herein. The processor **802** may comprise or be similar to the processors **118**, **202** discussed above.

(41) According to a fourth aspect, the present invention provides a personal care system. FIG. **9** is a schematic illustration of an example of a personal care system **900**. The personal care system **900** comprises a personal care device **104** having a treatment element **112** to be used to treat a portion of a user **102**. The personal care system **900** also comprises an image capture device **108** configured to capture image data in respect of the user of the personal care device **104**. The image capture device **108** may, for example, comprise a 2D or 3D camera. The personal care system **900** also comprises a display unit **106** (e.g. the display of an interactive mirror or of a computing device). The personal care system **900** also comprises processing apparatus **202**, **802**. The processing apparatus **118**, **202**, **802**, the display unit **106** and/or the image capture device **108** may form part of a single unit or component, such as a computing device or an interactive mirror. The personal care device **104**, the display unit **106** and the image capture device **108** may all be in communication with the processing apparatus **118**, **202**, **802**, for example via a wired or wireless connection.

(42) The processing apparatus **118**, **202**, **802** is configured to operate the image capture device **108** to capture image data in respect of the user **102** of the personal care device **104**, receive device operating data relating to the personal care device, determine position information of at least the treatment element **112** of the personal care device relative to the user, generate, based on the captured image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device, and provide the generated image for display on a representation of the user on the display unit.

(43) In some embodiments, the personal care device **104** may comprise a first sensor **114** configured to measure a force or pressure applied to the user **102** by the personal care device **104**. For example, the personal care device **104** may comprise a resistance sensor, a capacitance sensor and/or a piezoelectric sensor to measure a force applied to skin of the user **102** by the treatment element **112** of the personal care device during use. The processing apparatus **118**, **202**, **802** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** further based on an output of the first sensor **114**.

(44) In some embodiments, the personal care device **104** may comprise a second sensor **116** configured to measure orientation data and/or motion data in respect of the personal care device. For example, the personal care device **104** may comprise one or more of an accelerometer, and magnetometer, a gyroscope and an inertial measurement unit (IMU), configured to measure an orientation of the personal care device (e.g. relative to Earth) and/or data relating to any movement of the personal care device during use. The processing apparatus **118**, **202**, **802** may be configured to generate the image **120** representative of the predicted appearance of the portion of the user **102** further based on an output of the second sensor **116**.

(45) As noted above, the invention can be used in the field of personal care, or any field where a treatment is to be performed in respect of the user. In some embodiments, the personal care device **104** may comprise a device selected from a group comprising: a hair trimming device, a shaving device, a hair treatment device, a photoepilation device, a skin treatment device, a skin rejuvenation device and a product application device (e.g. a device for applying product such as cream, lotion, make up and the like) to a user.

(46) Embodiments of the invention disclosed herein provide a mechanism by which a user may view and anticipate the likely effect of performing a personal care activity before the activity is actually performed. This is achieved by predicting an appearance of part of the user if the personal care activity were performed, and displaying an image depicting the predicted appearance to the

user prior to the performance of the activity. In this way, a user may make an informed decision regarding whether or not to proceed with the personal care activity. Moreover, with foresight of the likely outcome of performing a personal care activity, the user may perform the personal care activity more accurately and quicker, reducing the power used by a personal care device, and reducing wear of components of the personal care device (e.g. the treatment element).

(47) The processor **118, 202, 802** can comprise one or more processors, processing units, multi-core processors or modules that are configured or programmed to control the apparatus **200** in the manner described herein. In particular implementations, the processor **118, 202, 802** can comprise a plurality of software and/or hardware modules that are each configured to perform, or are for performing, individual or multiple steps of the method described herein.

(48) The term “module”, as used herein is intended to include a hardware component, such as a processor or a component of a processor configured to perform a particular function, or a software component, such as a set of instruction data that has a particular function when executed by a processor.

(49) It will be appreciated that the embodiments of the invention also apply to computer programs, particularly computer programs on or in a carrier, adapted to put the invention into practice. The program may be in the form of a source code, an object code, a code intermediate source and an object code such as in a partially compiled form, or in any other form suitable for use in the implementation of the method according to embodiments of the invention. It will also be appreciated that such a program may have many different architectural designs. For example, a program code implementing the functionality of the method or system according to the invention may be sub-divided into one or more sub-routines. Many different ways of distributing the functionality among these sub-routines will be apparent to the skilled person. The sub-routines may be stored together in one executable file to form a self-contained program. Such an executable file may comprise computer-executable instructions, for example, processor instructions and/or interpreter instructions (e.g. Java interpreter instructions). Alternatively, one or more or all of the sub-routines may be stored in at least one external library file and linked with a main program either statically or dynamically, e.g. at run-time. The main program contains at least one call to at least one of the sub-routines. The sub-routines may also comprise function calls to each other. An embodiment relating to a computer program product comprises computer-executable instructions corresponding to each processing stage of at least one of the methods set forth herein. These instructions may be sub-divided into sub-routines and/or stored in one or more files that may be linked statically or dynamically. Another embodiment relating to a computer program product comprises computer-executable instructions corresponding to each means of at least one of the systems and/or products set forth herein. These instructions may be sub-divided into sub-routines and/or stored in one or more files that may be linked statically or dynamically.

(50) The carrier of a computer program may be any entity or device capable of carrying the program. For example, the carrier may include a data storage, such as a ROM, for example, a CD ROM or a semiconductor ROM, or a magnetic recording medium, for example, a hard disk. Furthermore, the carrier may be a transmissible carrier such as an electric or optical signal, which may be conveyed via electric or optical cable or by radio or other means. When the program is embodied in such a signal, the carrier may be constituted by such a cable or other device or means. Alternatively, the carrier may be an integrated circuit in which the program is embedded, the integrated circuit being adapted to perform, or used in the performance of, the relevant method.

(51) Variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the principles and techniques described herein, from a study of the drawings, the disclosure and the appended claims. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. A single processor or other unit may fulfil the functions of several items recited in the claims. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a

combination of these measures cannot be used to advantage. A computer program may be stored or distributed on a suitable medium, such as an optical storage medium or a solid-state medium supplied together with or as part of other hardware, but may also be distributed in other forms, such as via the Internet or other wired or wireless telecommunication systems. Any reference signs in the claims should not be construed as limiting the scope.

Claims

1. An apparatus comprising: a processor configured to receive image data in respect of a user of a personal care device, wherein the personal care device is to be used to treat a portion of the user; receive device operating data relating to the personal care device, wherein the device operating data comprise data indicative of an orientation and/or a motion of the personal care device; determine position information of at least a treatment element of the personal care device relative to the user; generate, based on the received image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; generate the image representative of the predicted appearance of the portion of the user having a size based on the orientation/motion data; and provide the generated image for display on a representation of the user.
2. The apparatus according to claim 1, wherein the processor is further configured to: determine, based on the received image data, a degree of deformation of a surface of the user by the personal care device; and generate the image representative of the predicted appearance of the portion of the user further based on the determined degree of deformation of the surface of the user.
3. The apparatus according to claim 1, wherein the processor is configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of a sensor of the personal care device.
4. The apparatus according to claim 1, wherein the processor is further configured to: determine, based on the received image data, an indication of the body part of the user that is to be treated using the personal care device; and generate the image representative of the predicted appearance of the portion of the user having a size based further on the indication of the body part.
5. The apparatus according to claim 1, wherein the apparatus comprises an interactive mirror, a desktop computer, a laptop computer, a tablet computer, a smartphone or a wearable device.
6. A computer-implemented method comprising: receiving image data in respect of a user of a personal care device, wherein the personal care device is to be used to treat a portion of the user; receiving device operating data relating to the personal care device, wherein the device operating data comprise data indicative of an orientation and/or a motion of the personal care device; determining position information of at least a treatment element of the personal care device relative to the user; generating, based on the received image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; generating the image representative of the predicted appearance of the portion of the user having a size based on the orientation/motion data; and display the generated image on a representation of the user.
7. The computer-implemented method according to claim 6, further comprising: displaying the representation of the user and the generated image on the representation of the user in an alternating manner.
8. The computer-implemented method according to claim 6, further comprising: determining, based on the received image data, an amount of deformation of a surface of the user by the personal care device wherein generating the image representative of the predicted appearance of the portion of the user is further based on the determined amount of deformation of the surface of the user.
9. The computer program product comprising a non-transitory computer-readable medium, the computer-readable medium having computer-readable code embodied therein, the computer-

readable code being configured such that, on execution by a suitable computer or processor, the computer or processor is caused to perform the method of claim 6.

10. A personal care system comprising: a personal care device having a treatment element to be used to treat a portion of a user; an image capture device configured to capture image data in respect of the user of the personal care device; a sensor configured to measure orientation data and/or motion data in respect of the personal care device; a display unit; and processing apparatus configured to: operate the image capture device to capture image data in respect of the user of the personal care device; receive device operating data relating to the personal care device; determine position information of at least the treatment element of the personal care device relative to the user; generate, based on the captured image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; and provide the generated image for display on a representation of the user on the display unit, wherein the processing apparatus is configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of the second sensor.

11. The personal care system according to claim 10, wherein the personal care device comprises: another sensor configured to measure a force or pressure applied to the user by the personal care device; and wherein the processing apparatus is configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of the other sensor.

12. The personal care system according to a claim 10, wherein the personal care device comprises a device selected from a group comprising: a hair trimming device, a shaving device, a hair treatment device, a photoepilation device, a skin treatment device, a skin rejuvenation device and a product application device.

13. An apparatus comprising: a processor configured to: receive image data in respect of a user of a personal care device, wherein the personal care device is to be used to treat a portion of the user; receive device operating data relating to the personal care device; determine position information of at least a treatment element of the personal care device relative to the user; generate, based on the received image data, the received device operating data and the determined position information, an image representative of a predicted appearance of the portion of the user following treatment using the personal care device; and provide the generated image for display on a representation of the user, wherein the device operating data comprises a size of the treatment element of the personal care device; generate the image representative of the predicted appearance of the portion of the user having a size based on the size of the treatment element, wherein the device operating data comprises a size of the treatment element of the personal care device; and generate the image representative of the predicted appearance of the portion of the user having a size based on the size of the treatment element.

14. The apparatus according to claim 13, wherein the processor is further configured to: determine, based on the received image data, a degree of deformation of a surface of the user by the personal care device; and generate the image representative of the predicted appearance of the portion of the user further based on the determined degree of deformation of the surface of the user.

15. The apparatus according to claim 13, wherein the processor is further configured to generate the image representative of the predicted appearance of the portion of the user further based on an output of a sensor of the personal care device.

16. The apparatus according to claim 13, wherein the device operating data comprise data indicative of an orientation and/or a motion of the personal care device, wherein the processor is configured to generate the image representative of the predicted appearance of the portion of the user having a size based on the orientation/motion data.

17. The apparatus according to claim 13, wherein the processor is further configured to: determine, based on the received image data, an indication of the body part of the user that is to be treated using the personal care device; and generate the image representative of the predicted appearance

of the portion of the user having a size based further on the indication of the body part.

18. An apparatus according to claim 13, wherein the apparatus comprises an interactive mirror, a desktop computer, a laptop computer, a tablet computer, a smartphone or a wearable device.
